EuroMCM 2026 Problem ɣ : Tell Me, Muse



Ulysses and the Sirens by John William Waterhouse, 1891

1 Track Notes

Problem ɣ is the Big Data problem, though it also touches on the content of the Operations Research/ Network Science (δ) track. The track label describes the problem, not the methods a team is expected to use: solutions that draw on methods associated with any EuroMCM track are equally acceptable, and the choice of approach has no bearing on how a paper is judged. Creative solutions are encouraged.

2 Background

The *Odyssey* tells of *Odysseus*' ten-year journey home from the Trojan War. With the *Iliad*, it stands at the beginning of European literature. Both poems grew out of a long tradition of oral performance, and both are attributed to Homer, a poet of whom almost nothing is known.

Whether one poet composed them has been argued for more than two thousand years. Some ancient scholars held that the two poems had different authors. Two of the most respected, Aristophanes of Byzantium and Aristarchus, held that the *Odyssey* ends at Book 23, line 296, when *Odysseus* and Penelope go to bed, and that the last 600 lines were added later. Modern readers have also asked whether the first four books, which follow *Odysseus*' son Telemachus rather than *Odysseus* himself, were once a separate poem.

The editors of a digital library of classical texts are preparing new online editions of both poems. They must decide whether to mark any part of either poem as being of disputed authorship. They have asked your team for an analysis based on the text itself.

3 Requirements

Your solution should explore and address the following aspects:

- Develop a model that characterises the style of Greek epic verse. How well does it tell the different poets in the data apart?
- Use your model to assess the ancient claim that the *Odyssey* originally ended at Book 23, line 296.

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- One poet or several? Assess whether the *Iliad* and the *Odyssey* were composed by the same poet, and whether the Telemachus books (*Odyssey* 1–4) were once a separate poem. Do other parts of either poem stand out?
- The poem often states how long a voyage took. Develop a model that relates sailing time to distance, and use it to estimate where the stops in *Odysseus*' wanderings could lie. Can the route in Books 9–12 be placed on a real map?
- Books 9–12 are told by *Odysseus* himself, not by the narrator of the poem. Discuss how this bears on your findings about style and about geography.

4 Share Your Insights

Write a one-page memo to the editorial board. Recommend whether to mark any passage as disputed and how to show *Odysseus*' route on their map, and say how strongly the evidence supports each recommendation.

5 Data

The accompanying dataset, EuroMCM_2026_Problem_Gamma_Datasets.zip, is available from the 2026 Problems page on the official EuroMCM website.

Primary

- `Odyssey_Greek_English.csv` and `Iliad_Greek_English.csv`: each row is a passage of about five lines, with book, first line, last line, the Greek text and the English prose translation by A. T. Murray.
- `Odyssey_Greek_Lines.csv` and `Iliad_Greek_Lines.csv`: each row is one line of Greek verse.

Comparison texts

Each row is one line of Greek verse. None of these was composed by the poet of the *Iliad* or the *Odyssey*.

- `Hesiod_Greek_Lines.csv`: three poems traditionally attributed to Hesiod, an early poet in the same metre.
- `Homeric_Hymns_Greek_Lines.csv`: 33 hymns to the gods, attributed to Homer in antiquity but composed by several later poets.
- `Argonautica_Greek_Lines.csv`: the *Argonautica* of Apollonius of Rhodes, an epic written about five centuries after Homer in deliberate imitation of him.

Places

- `Odyssey_Places.csv`: coordinates of 22 real places named on voyages in the *Odyssey*, from the Pleiades gazetteer of ancient places. Places that may be imaginary are left out on purpose.

No knowledge of Ancient Greek is required. The data do not cover winds, currents or the ships of the period, which you may need to take into account. There is no restriction on incorporating external data to augment and validate the solution, provided sources and assumptions are stated clearly.

6 Citation

Please cite the source datasets:

```
Perseus Digital Library. canonical-greekLit [Data set]. GitHub. https://github.com/
PerseusDL/canonical-greekLit. Licensed under CC BY-SA 4.0.

Bagnall, R., et al. (Eds.). Pleiades: A Gazetteer of Past Places. https://pleiades.
stoa.org. Licensed under CC BY 3.0.
```

The data files were derived from these sources, with changes to their format and structure described in the README. The Perseus-derived text files are distributed under CC BY-SA 4.0. The Pleiades-derived place data are distributed under CC BY 3.0.

7 Submission

Your PDF solution (≤25 pages) should include:

- One-page Summary Sheet
- Table of Contents
- Complete Solution
- One-page Memo
- References
- Report on Use of AI Tools (if applicable; excluded from the 25-page limit)

There is no mandatory minimum length. Teams may submit incomplete solutions. The use of AI tools is allowed but optional; compliance with EuroMCM AI usage policy is required.

8 Links

J. P. Emanuel, *Odysseus*' raid on Egypt in its Late Bronze Age context (2012)

P. Gheorghiade and C. Spencer, Modelling the cost of the wind: networks of mobility in the Late Bronze Age Mediterranean (2024)

C. Knappett, T. Evans and R. Rivers, Modelling maritime interaction in the Aegean Bronze Age (2008)

9 Glossary

Dactylic hexameter: The six-beat metre of Greek epic. Every line of every poem in the data is written in it.

Formula: A phrase repeated in fixed form to fill part of a line, such as "rosy-fingered Dawn". Formulas are characteristic of poetry composed in oral performance.